

1EME Submission

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1 Summary

We estimate pandemic-era excess mortality for Maryland and Kentucky using the `excessmort` package in R. The core approach fits an overdispersed Poisson regression model to pre-pandemic monthly death counts, incorporating a slow secular time trend (via spline) and seasonal variation (two Fourier harmonics at monthly frequency). Expected death counts under a counterfactual no-pandemic scenario were then extrapolated across the pandemic interval (March 1918 – May 1920). We estimate deaths and death rates along with associated 95% confidence intervals. The **Estimates** section displays these.

Pandemic deaths were defined conservatively across estimands 1 and 3 as deaths attributed to influenza, pulmonary tuberculosis, or bronchitis — causes that are directly associated with the 1918 pandemic. The three estimands differ in their unit of analysis and stratification.

Estimand 1: State-level: Pandemic-cause deaths were aggregated to the state-month level across all counties and racial groups, and a single excess model was fit for each state.

Estimand 2: County-level: The excess model was run independently for each county using pandemic-cause deaths, allowing local variation in mortality burden to be captured. For estimand 2, we rely on granular death count files, which do not have cause attributed to them.

Estimand 3: State-level by race: Pandemic-cause deaths were aggregated to the state-month level separately by racial group, and excess models were fit independently for each race stratum. This estimand examines whether the pandemic’s mortality burden fell disproportionately across racial groups. Population denominators were interpolated linearly between decadal census years within each county.

2 Estimates

Table 1: Estimand 1: The total number and rate of pandemic deaths in Maryland and Kentucky

	Kentucky	Maryland
Pandemic Death Rate (95 percent CI)	89.72 [51.01, 128.44]	19.6 [14.6, 24.6]
Pandemic Death Number (95 percent CI)	107874 [61329.09, 154418.91]	5838 [4349.02, 7326.98]

Table 2: Estimand 2: The total number and rate of panedmic deaths in Maryland

County	Pandemic Death Rate (95 percent CI)	Pandemic Death Number (95 percent CI)
Allegany County	0.45 [0.34, 0.56]	139 [105.47, 172.53]
Calvert County	2.85 [1.82, 3.89]	28 [17.83, 38.17]
Caroline County	1.31 [0.93, 1.69]	50 [35.54, 64.46]
Carroll County	0.85 [0.67, 1.02]	102 [81.3, 122.7]
Cecil County	0.98 [0.72, 1.24]	57 [41.82, 72.18]
Charles County	1.84 [1.27, 2.4]	37 [25.62, 48.38]
Dorchester County	1.22 [0.94, 1.49]	72 [55.6, 88.4]
Frederick County	0.72 [0.61, 0.83]	153 [128.99, 177.01]
Garrett County	0.93 [0.68, 1.17]	47 [34.31, 59.69]
Harford County	0.88 [0.66, 1.11]	61 [45.22, 76.78]
Howard County	1.35 [0.84, 1.86]	32 [19.9, 44.1]
Kent County	1.78 [1.19, 2.36]	39 [26.14, 51.86]
Montgomery County	0.86 [0.66, 1.07]	68 [51.79, 84.21]
Prince George's County	0.76 [0.59, 0.92]	84 [65.73, 102.27]
Queen Anne's County	1.61 [0.98, 2.23]	28 [17.09, 38.91]
Somerset County	1.85 [1.38, 2.32]	62 [46.28, 77.72]
St. Mary's County	1.42 [1.01, 1.83]	39 [27.73, 50.27]
Talbot County	1.56 [1.15, 1.97]	56 [41.29, 70.71]
Washington County	0.52 [0.42, 0.62]	128 [103.18, 152.82]
Wicomico County	1.08 [0.86, 1.3]	89 [70.53, 107.47]
Worcester County	1.52 [1.15, 1.88]	58 [43.99, 72.01]

Table 3: Estimand 2: The total number and rate of panedmic deaths in Kentucky

County	Pandemic Death Rate (95 percent CI)	Pandemic Death Number (95 percent CI)
Adair County	0.97 [0.65, 1.29]	32 [21.3, 42.7]
Allen County	1.07 [0.74, 1.4]	37 [25.64, 48.36]
Anderson County	1.48 [0.87, 2.1]	21 [12.27, 29.73]
Ballard County	1.14 [0.66, 1.62]	23 [13.35, 32.65]
Barren County	0.81 [0.63, 0.99]	70 [54.21, 85.79]
Bath County	1.34 [0.85, 1.83]	26 [16.47, 35.53]
Bell County	0.51 [0.36, 0.65]	51 [36.31, 65.69]
Boone County	1.52 [0.91, 2.13]	23 [13.7, 32.3]
Bourbon County	0.83 [0.59, 1.07]	52 [36.74, 67.26]
Boyd County	0.71 [0.55, 0.88]	68 [52.4, 83.6]
Boyle County	1 [0.66, 1.33]	36 [23.82, 48.18]
Bracken County	1.26 [0.58, 1.93]	15 [6.95, 23.05]
Breathitt County	0.67 [0.35, 1]	18 [9.44, 26.56]
Breckinridge County	0.83 [0.61, 1.06]	51 [37.14, 64.86]
Bullitt County	1.5 [0.85, 2.15]	21 [11.87, 30.13]
Butler County	0.97 [0.63, 1.32]	31 [20.07, 41.93]
Caldwell County	0.95 [0.51, 1.38]	21 [11.35, 30.65]
Calloway County	0.86 [0.59, 1.12]	34 [23.39, 44.61]
Campbell County	0.44 [0.36, 0.53]	116 [94.46, 137.54]
Carlisle County	1.64 [0.97, 2.31]	24 [14.16, 33.84]
Carroll County	1.53 [0.78, 2.28]	18 [9.14, 26.86]
Carter County	0.79 [0.58, 1]	53 [38.88, 67.12]
Casey County	0.8 [0.4, 1.21]	16 [7.89, 24.11]
Christian County	0.7 [0.58, 0.83]	125 [102.84, 147.16]
Clark County	1.25 [0.97, 1.54]	60 [46.33, 73.67]
Clay County	0.81 [0.38, 1.25]	12 [5.6, 18.4]
Clinton County	1.4 [0.01, 2.79]	4 [0.02, 7.98]
Crittenden County	1.28 [0.77, 1.78]	21 [12.69, 29.31]
Cumberland County	1.14 [0.48, 1.8]	13 [5.45, 20.55]
Daviess County	0.6 [0.49, 0.71]	104 [84.25, 123.75]
Edmonson County	1.18 [0.55, 1.82]	15 [7.02, 22.98]
Elliott County	1.42 [0.71, 2.13]	16 [7.96, 24.04]
Estill County	1.04 [0.61, 1.46]	21 [12.34, 29.66]
Fayette County	0.75 [0.63, 0.87]	196 [163.77, 228.23]
Fleming County	0.96 [0.65, 1.28]	39 [26.26, 51.74]
Floyd County	0.5 [0.28, 0.72]	22 [12.32, 31.68]
Franklin County	1 [0.76, 1.25]	50 [44.76, 73.24]
Fulton County	1.27 [0.88, 1.65]	35 [24.28, 45.72]
Gallatin County	2.94 [0.92, 4.95]	8 [2.5, 13.5]
Garrard County	1.2 [0.75, 1.66]	25 [15.59, 34.41]
Grant County	1.26 [0.71, 1.81]	22 [12.39, 31.61]
Graves County	0.66 [0.52, 0.8]	77 [60.51, 93.49]
Grayson County	0.8 [0.53, 1.08]	32 [21.07, 42.93]
Green County	1.31 [0.75, 1.86]	20 [11.53, 28.47]
Greene County	0.87 [0.62, 1.11]	43 [30.66, 55.34]
Hancock County	1.82 [0.82, 2.82]	13 [5.85, 20.15]
Hardin County	0.68 [0.47, 0.89]	41 [28.37, 53.63]
Harlan County	0.65 [0.49, 0.81]	56 [41.86, 70.14]
Harrison County	0.87 [0.53, 1.21]	29 [17.7, 40.3]
Hart County	0.81 [0.51, 1.11]	30 [19.04, 40.96]
Henderson County	0.71 [0.55, 0.88]	73 [56.08, 89.92]
Henry County	1.12 [0.74, 1.49]	35 [23.33, 46.67]
Hickman County	1.15 [0.52, 1.78]	16 [7.21, 24.79]
Hopkins County	0.55 [0.4, 0.7]	58 [42.12, 73.88]
Jackson County	1.2 [0.39, 1.81]	15 [7.35, 22.65]
Jefferson County	0.54 [0.47, 0.62]	796 [683.69, 908.31]
Jessamine County	1.25 [0.75, 1.75]	23 [13.75, 32.25]
Johnson County	0.93 [0.63, 1.23]	30 [20.31, 39.69]
Kenton County	0.56 [0.48, 0.65]	206 [174.6, 237.4]
Knott County	1.21 [0.7, 1.73]	21 [12.1, 29.9]
Knox County	0.72 [0.49, 0.94]	40 [27.36, 52.64]
Laurel County	0.92 [0.65, 1.2]	35 [24.52, 45.48]
Lawrence County	0.9 [0.57, 1.23]	27 [17.12, 36.88]
Lee County	1.46 [0.47, 2.44]	7 [2.28, 11.72]
Leslie County	1.29 [0.64, 1.93]	16 [8, 24]
Letcher County	0.61 [0.37, 0.85]	29 [17.7, 40.3]
Lewis County	1.11 [0.82, 1.39]	53 [39.42, 66.58]
Lincoln County	1.06 [0.79, 1.34]	53 [39.25, 66.75]
Livingston County	1.36 [0.75, 1.97]	20 [11.02, 28.98]
Logan County	0.78 [0.58, 0.98]	54 [39.92, 68.08]
Lyon County	1.44 [0.71, 2.18]	16 [7.85, 24.15]
M-P-W Counties	0.34 [0.26, 0.42]	146 [112.03, 179.97]
Madison County	0.72 [0.54, 0.89]	60 [45.31, 74.69]
Magoffin County	1.16 [0.63, 1.69]	16 [8.66, 23.34]
Marion County	1.17 [0.84, 1.49]	44 [31.89, 56.11]
Marshall County	0.92 [0.56, 1.28]	27 [16.52, 37.48]
Martin County	1.58 [0.38, 2.77]	7 [1.7, 12.3]
Mason County	0.94 [0.67, 1.21]	46 [32.76, 59.24]
McCracken County	0.75 [0.62, 0.89]	102 [83.68, 120.32]
McLean County	1.19 [0.73, 1.65]	25 [15.28, 34.72]
Meade County	1.45 [0.84, 2.07]	23 [13.35, 32.65]
Menifee County	2.58 [1.11, 4.04]	10 [4.3, 15.7]
Mercer County	1.18 [0.83, 1.52]	39 [27.58, 50.42]
Metcalfe County	1.37 [0.67, 2.07]	15 [7.32, 22.68]
Monroe County	0.97 [0.56, 1.39]	23 [13.28, 32.72]
Montgomery County	1.22 [0.78, 1.65]	30 [19.31, 40.69]
Morgan County	0.94 [0.59, 1.29]	27 [16.93, 37.07]
Muhlenberg County	0.69 [0.55, 0.84]	74 [58.18, 89.82]
Nelson County	0.98 [0.69, 1.27]	45 [31.71, 58.29]
Nicholas County	1.5 [0.92, 2.08]	25 [15.35, 34.65]
Ohio County	0.73 [0.55, 0.91]	55 [41.26, 68.74]
Oldham County	1.57 [0.64, 2.5]	13 [5.29, 20.71]
Owen County	1.11 [0.68, 1.54]	27 [16.56, 37.44]
Owsley County	1.7 [0.71, 2.68]	11 [4.6, 17.4]
Pendleton County	1.13 [0.61, 1.66]	20 [10.73, 29.27]
Perry County	0.7 [0.5, 0.89]	48 [34.6, 61.4]
Pike County	0.48 [0.38, 0.58]	84 [66.76, 101.24]
Powell County	2.31 [0.91, 3.71]	9 [3.54, 14.46]
Robertson County	3.08 [0.54, 5.62]	6 [1.06, 10.94]
Rockcastle County	1.01 [0.64, 1.37]	27 [17.25, 36.75]
Rowan County	1.38 [0.81, 1.95]	24 [14.11, 33.89]
Russell County	1.07 [0.6, 1.54]	22 [12.29, 31.71]
Scott County	1.05 [0.76, 1.34]	49 [35.61, 62.39]
Shelby County	0.83 [0.57, 1.1]	40 [27.28, 52.72]
Simpson County	1.12 [0.62, 1.63]	23 [12.67, 33.33]
Spencer County	1.72 [0.94, 2.49]	20 [10.97, 29.03]
Taylor County	1.15 [0.64, 1.65]	21 [11.78, 30.22]
Todd County	0.86 [0.55, 1.17]	34 [21.64, 46.36]
Trigg County	0.93 [0.51, 1.34]	22 [12.19, 31.81]
Trimble County	2.19 [0.86, 3.53]	10 [3.9, 16.1]
Union County	0.85 [0.54, 1.17]	30 [19.03, 40.97]
Warren County	0.66 [0.51, 0.81]	71 [54.5, 87.5]
Washington County	0.91 [0.49, 1.33]	20 [10.79, 29.21]
Webster County	0.85 [0.6, 1.09]	47 [33.5, 60.5]
Wolfe County	1.34 [0.63, 2.05]	16 [7.55, 24.45]
Woodford County	1.33 [0.89, 1.78]	34 [22.65, 45.35]

Table 4: Estimand 3: The total number and rate of pandemic deaths by race in Maryland

Race	Pandemic Death Rate (95 percent CI)	Pandemic Death Number (95 percent CI)
Non-white	34.04 [29.3, 38.78]	10458 [9001.81, 11914.19]
White	21.62 [18.75, 24.5]	27699 [24018.52, 31379.48]